INCH-POUND

MIL-S-20708/38
13 March 1989

SUPERSEDING
MIL-S-20708/38
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO, TORQUE RECEIVER, TYPE 11TR4C

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708

TABLE I Requirements

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	115	volts	±1%
Primary Current	60 0	milliamps	maximum
Primary Power	1 0	watts	maximum
Impedance Zss	180-250	ohms	min -max
Impedance Angle Zss	22 0 28 0	degrees	min -max
Transformation Ratio	0 783	------	±2%
Phase Shift (Lead)	6 0	degrees	maximum
Electrical Error	----	minutes	maximum
Receiver Error	60 0	minutes	maximum
Synchronizing Time 30° ±2°	1 0	seconds	maximum
Synchronizing Time 177° ±2°	2 0	seconds	maximum
Torque Gradient	0 0080	ounce-inches per degree	minimum
Radial Play	0 0006	inches	maximum
End Play	0 0025	inches	±0 0020
Temperature Rise	25 0	degrees (C)	maximum
Variation of Voltage (+10%) and Freq (-5%)	1 45	watts	maximum

(E) denotes changes

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TABLE II Military part number variant characteristics

Military Part No 1/	A ± 010	B ± 010	G Maximum	Shaft Style	Terminal End	U + 0000 - 0002
M20708/3-01C	555	500	1 732	Splined	Terminal Board	1872
M20708/3-02C	368	312	1 630	Plain & Slotted	Wire Leads	1200
M20708/3-03C	430	375	↓	↓	↓	↓
M20708/3-04C	493	437				
M20708/3-05C	555	500				
M20708/3-06C	618	562				

1/ Military part number M20708/3-01C shall be the NATO standard

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation

TABLE III Military part number cross-references

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBERS
MIL-S-20708/3C	MIL-S-20708/3B	
M20708/3C-001	M20708/3-001	M20708/3-01C
M20708/3C-002		M20708/3-02C
M20708/3C-003		M20708/3-03C
M20708/3C-004		M20708/3-04C
M20708/3C-005		M20708/3-05C
M20708/3C-006		M20708/3-06C

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification Part number changes do not affect form, fit, or function of the synchros listed therein

Custodians
Army-AR
Navy-AS
Air Force-85

Review Activities
Army-MI,AV
Navy-OS
Air Force-11 17,99
DLA-ES

User Activities
Navy-MC,CG

Preparing Activity
Navy-AS

(Project 5990-0349-3)

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